

Reflective Summary – Distributed Machine Learning on Banking Data

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1. Project Overview

In this project, I implemented a distributed banking analytics and machine learning system using Hadoop HDFS, Hive, Apache Spark, Spark ML, Spark Structured Streaming, and Python. The objective was to simulate a real-world banking analytics environment capable of processing large-scale data efficiently using distributed computing technologies.

2. Challenges Faced

- One of the biggest challenges during the project was configuring Hadoop, Hive, and Spark on Windows 11. Several environment compatibility issues occurred while connecting Java, Hadoop, Spark, and PySpark together.
- Another major challenge was setting up Spark sessions properly inside Jupyter Notebook. Initially, Python version compatibility problems caused Spark initialization errors, which were later resolved by switching to Python 3.11.
- Streaming implementation was also challenging because Spark Structured Streaming requires proper schema handling, checkpointing, and continuous input monitoring.
- Hadoop Streaming execution on Windows caused multiple issues and instability. To overcome this problem, a local MapReduce simulation using Python mapper and reducer scripts was implemented successfully.
- Managing multiple services such as NameNode, DataNode, Hive, Spark, and Jupyter simultaneously also required careful resource handling and troubleshooting.

3. Learning Outcomes

This project provided practical exposure to distributed computing systems and large-scale analytics.

Major learning outcomes include:

- Understanding distributed storage using HDFS
- Understanding Spark distributed processing architecture
- Implementing distributed SQL analytics using Hive and Spark SQL

- Building distributed ML pipelines using Spark ML
- Performing feature engineering
- Implementing real-time streaming analytics
- Understanding data parallelism and partitioning
- Learning watermarking and window operations in streaming systems
- Understanding the MapReduce workflow
- Managing large-scale datasets using distributed frameworks

4. Practical Understanding Gained

This project helped in understanding how modern banking systems process large amounts of customer and transaction data efficiently. It demonstrated the importance of distributed systems for scalability, fault tolerance, and real-time analytics.

The project also provided insight into how banks use machine learning for customer behavior prediction, campaign optimization, and real-time monitoring.

Implementing Spark ML and Structured Streaming gave practical experience in scalable machine learning and streaming architectures.

5. Skills Improved

The project improved both technical and problem-solving skills.

Technical skills improved:

- Hadoop and HDFS
- HiveQL
- PySpark
- Spark ML
- Spark Streaming
- Distributed analytics
- Machine learning pipelines

Soft skills improved:

- Debugging, Research ability, System troubleshooting

6. Conclusion

Overall, this project provided strong practical exposure to distributed computing and machine learning systems. It demonstrated how big data technologies can be combined to build scalable analytics systems for banking environments.